Ratios, rates and proportions

Day 13 • 2026-11-19 • 12:45 PM–2:00 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

Before checking a ratio in the Numeracy Hub calculator, what must be made consistent?

A. The number of decimal digits only

B. The font size of the labels

C. The colour of each item

D. The units of both quantities



Numeracy Hub

Choice (A–D) _____ **Working / reason** _____

Textbook sections 6.5–6.7 pp.144–148

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02 Which ratio is 12:18 in lowest whole-number terms?

A. 2:3

B. 6:9

C. 3:2

D. 1:6

Choice (A–D)

Working / reason

Textbook sections 6.5–6.7 pp.144–148

03 Which ratio is equivalent to 3:5?

A. 3:10

B. 6:5

C. 6:10

D. 5:3

Choice (A–D)

Working / reason

Textbook sections 6.5–6.7 pp.144–148

04 What does a rate of 4 boards per hour describe?

A. Four boards completed in each hour at that rate

B. A dimensionless ratio only

C. Four hours for every board

D. Four boards total regardless of time

Choice (A–D)

Working / reason

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05 If the number of identical frames doubles while the material per frame stays fixed, what happens to total material?

A. It increases by exactly one unit
B. It halves
C. It stays the same
D. It doubles

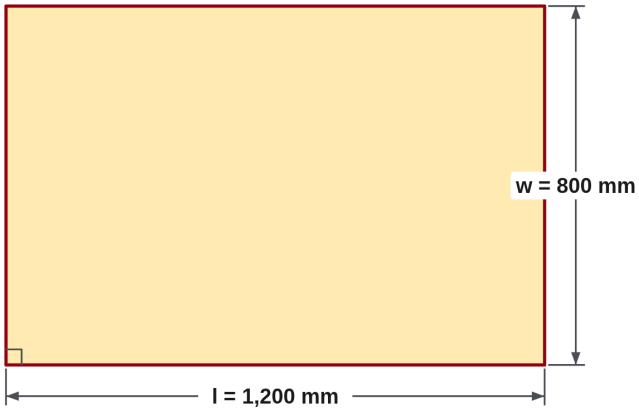
Choice (A–D)

Working / reason

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06 **CALCULATE**

The pictured rectangle is 1200 mm by 800 mm. Its long-to-short ratio is 3:2. If the short side becomes 1000 mm at the same ratio, find the long side.



Shape Canvas

Answer (mm)

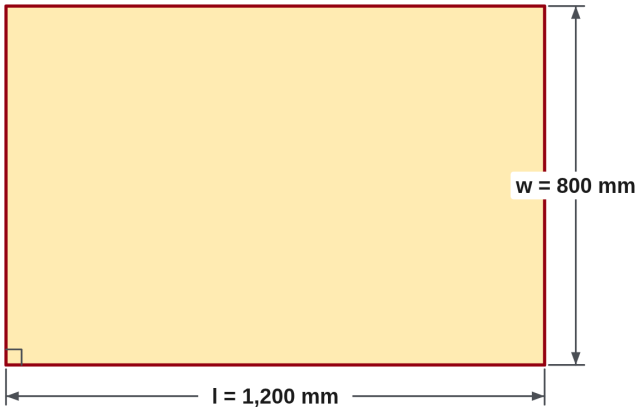
Working / reason

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07 The pictured rectangle is 1200 mm by 800 mm. Scale both sides by 1.5. Find the new short side.



Shape Canvas

Answer (mm) **Working / reason**

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08 Three identical packs contain 72 screws altogether. How many screws are in one pack?

Answer (screws) **Working / reason**

Textbook sections 6.5–6.7 pp.144–148

09 A worker completes 18 equal pieces in 3 hours. At the same rate, how many pieces are completed in 5 hours?

Answer (pieces) **Working / reason**

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10 A mix uses A:B = 2:3. If 8 kg of A is used, how much B is required?

Answer (kg) | **Working / reason**

Textbook sections 6.5–6.7 pp.144–148

11 APPLY & CHECK

Four identical frames require 18 m of material. At the same rate, how many metres are needed for ten frames?

Answer (m) | **Working / reason**

Textbook sections 6.5–6.7 pp.144–148

12 A mix is A:B = 3:2. The total mix is 35 kg. Find the mass of B.

Answer (kg) | **Working / reason**

Textbook sections 6.5–6.7 pp.144–148

13 A learner simplifies 1200 mm:800 mm to 2:3. What is the correction?

A. Ratios cannot be simplified
B. The correct ratio is 1:1
C. The long-to-short ratio is 3:2
D. The correct ratio is 12:80

Choice (A–D) | **Working / reason**

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14 At a constant rate, 7 m of trim costs $29.40. What do 12 m cost?

Answer ($)

Working / reason

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15 Which check supports a proportional material estimate?

A. Only the final numbers have equal digits

B. The larger order always uses less material

C. The quantity per item remains the same

D. Every quantity is rounded down first

Choice (A–D)

Working / reason

Textbook sections 6.5–6.7 pp.144–148